

Smart City Traffic Intelligence — Part 3 Machine Learning and AI Capstone Report

Then add this first section:

1. Supervised Machine Learning

A common feature set was developed using time-based variables, cyclical hour and day-of-week encodings, weekday/weekend indicators, holiday status, weather measurements and one-hot encoded weather categories. Two supervised-learning problems were evaluated: classification of a proxy high-risk traffic condition and regression prediction of traffic volume.

For classification, the `high_risk` target was created because no real accident dataset was provided. A record was labelled high risk when High or Severe congestion occurred together with severe or low-visibility weather. Logistic Regression and Random Forest Classifier were compared. Logistic Regression achieved an accuracy of 96.53%, precision of 77.36%, recall of 97.98%, F1-score of 86.46%, and ROC AUC of 0.9937. Random Forest achieved an accuracy of 98.38%, precision of 92.52%, recall of 93.20%, F1-score of 92.86%, and ROC AUC of 0.9965. Random Forest provided stronger overall performance, while Logistic Regression achieved higher recall and therefore detected a larger proportion of proxy high-risk cases.

For traffic-volume regression, Linear Regression and Random Forest Regressor were compared. Linear Regression achieved an MAE of 813.47 vehicles and an R^2 of 0.7260. Random Forest Regressor achieved a substantially lower MAE of 271.26 vehicles and a higher R^2 of 0.9442, making it the preferred regression model.

2. Unsupervised Machine Learning

K-means clustering and association rule mining were applied to identify traffic patterns without using a predefined target variable.

For K-means clustering, three features were used: Hour, weather severity, and traffic volume. The variables were standardised before clustering so that traffic volume would not dominate because of its larger numerical scale. A three-cluster solution was used.

The clusters were interpreted as follows:

- Cluster 0 — Low-Traffic Overnight: average hour 2.93, average weather severity 1.81, and average traffic volume 816.25 vehicles.
- Cluster 1 — High-Traffic Mild Weather: average hour 15.12, average weather severity 0.65, and average traffic volume 4,254.11 vehicles.
- Cluster 2 — High-Traffic Adverse Weather: average hour 13.79, average weather severity 3.78, and average traffic volume 4,091.23 vehicles.

These results suggest that time of day and weather conditions help distinguish meaningful traffic-condition groups.

Association rule mining was then used to identify combinations of time of day, weekday type, weather, and congestion level. One of the strongest rules was Overnight + Weekend → Low Congestion, with confidence of 88.61% and lift of 3.54. This indicates that low congestion was substantially more likely during weekend overnight periods than overall. Another important rule was Weekday + Afternoon → Severe Congestion, with confidence of 71.33% and lift of 2.86. This suggests that weekday afternoons were strongly associated with severe congestion.

Overall, the unsupervised-learning analysis revealed interpretable mobility patterns that complement the supervised prediction models.

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4. Advanced AI Technique — MLflow Experiment Tracking

MLflow was selected as the advanced AI technique because it supports experiment tracking and also connects directly to the MLOps requirements in Task 6. A local SQLite backend was used to store experiment metadata, while a shorter local artifact directory was used for model artifacts to avoid Windows path-length limitations.

Three regression experiments were tracked: Linear Regression, Random Forest Regressor, and Neural Network. MLflow recorded model parameters, performance metrics and run information, allowing the models to be compared consistently.

The tracked results were:

- Linear Regression: MAE 813.47, R^2 0.7260
- Random Forest Regressor: MAE 271.26, R^2 0.9442
- Neural Network: MAE 294.53, R^2 0.9403

The Random Forest Regressor achieved the best overall regression performance and was therefore selected for deployment.

MLflow adds value by creating a structured and reproducible history of model experiments. It makes it easier to compare model versions, review parameter settings and identify which model should be promoted for deployment. However, MLflow also introduces additional setup and storage requirements. In this project, configuration adjustments were required because newer MLflow versions no longer use the filesystem tracking backend by default, and Windows path-length limitations affected model artifact storage

5. Traffic Recommendation System

A travel-timing recommendation system was developed to translate the traffic analysis into practical mobility guidance. Because the dataset represents a single traffic corridor rather than multiple alternative routes, the system focuses on recommending when to travel instead of which route to take.

Historical traffic volume was analysed by day type, hour and weather condition. To make the recommendations more practical, travel times were restricted to the period from 06:00 to 22:00. Weather-specific recommendations were only generated when at least 20 historical observations were available for a given day-type, weather and hour combination. This minimum-sample rule was used to reduce the risk of generating recommendations from very limited historical evidence.

The analysis showed that quieter weekday travel periods tended to occur later in the evening, while quieter weekend periods tended to occur earlier in the morning. For example, under weekday and clear-weather conditions, the system recommended

travelling between 22:00 and 23:00, when historical traffic volume averaged approximately 2,223 vehicles.

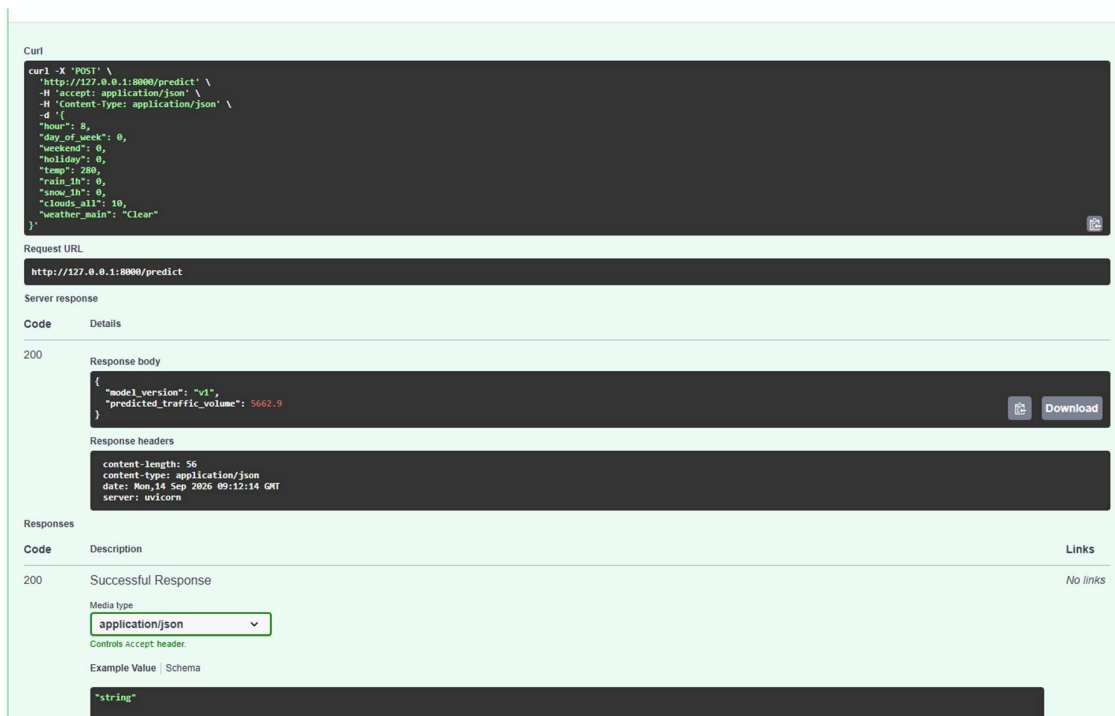
The recommendation engine was also implemented as a reusable Python script. It loads the processed Part 2 dataset, constructs reliable travel profiles, and generates a plain-language recommendation based on the selected day type and weather condition. The system is intended as decision support based on historical patterns rather than a guarantee of future traffic conditions.

6. MLOps, Deployment and Monitoring

The project simulated how the traffic-prediction model could move from experimentation into deployment. Model versions were documented as v0 Linear Regression, v1 Random Forest Regressor, and v2 Neural Network. Based on model performance, the Random Forest Regressor was selected for deployment because it achieved the lowest MAE (271.26) and the highest R^2 (0.9442).

MLflow was used for experiment tracking, recording model parameters, metrics and experiment runs. The selected Random Forest model was also saved as a reusable joblib model bundle containing the trained model, required feature columns, version information and performance metrics.

A FastAPI deployment mock-up was created to serve the Random Forest model. The API accepts traffic-related inputs such as hour, day of week, holiday status, temperature, rainfall, cloud cover and weather condition. These inputs are converted into the same engineered feature structure used during model training before the model returns a predicted traffic volume. Testing through FastAPI's interactive documentation returned a successful HTTP 200 response, demonstrating that the deployed model could accept input and generate predictions.



Model monitoring was simulated by comparing current prediction error against the original deployment MAE. A monitoring threshold was set at 20% above the baseline MAE. The baseline MAE was 271.26, producing an alert threshold of 325.51. The monitored MAE remained at 271.26, resulting in a PASS / Normal status. A simulated future MAE of 350 exceeded the threshold and correctly produced an ALERT / Requires investigation status

7. Responsible and Sustainable AI

Responsible AI considerations were evaluated across bias, fairness, governance and sustainability. The dataset has several coverage limitations, including incomplete observations across some years and representation of only a single traffic corridor. These limitations may reduce the model's ability to generalise to other locations, seasons or future traffic patterns.

Because no real accident dataset was provided, a proxy high_risk label was created using congestion and adverse-weather conditions. This proxy is suitable for demonstrating the classification workflow but should not be interpreted as a real accident-prediction target. Strong classification performance may partly result from the close relationship between the features and the rules used to construct the proxy label. Model errors may also differ across conditions such as time of day, weekday versus weekend and rare weather categories.

For governance in mobility AI, predictions and travel recommendations should be treated as decision-support information, not automatic instructions. Traffic planners or

transport operations staff should review model performance, limitations and monitoring alerts before using the system for real-world mobility decisions. Model updates should be validated and approved before deployment, with clear responsibility for investigating alerts and unexpected predictions.

Sustainability should also influence modelling choices. In this project, the Random Forest Regressor achieved slightly better performance than the neural network while requiring a simpler training and deployment process. Selecting a model that provides strong performance without unnecessary computational complexity can reduce processing time, energy use and infrastructure requirements. In larger mobility systems, the resource costs of sensor collection, cloud storage, model retraining and continuous inference should also be considered.

A more detailed discussion is provided in the separate Responsible AI: Bias, Fairness, Governance and Sustainability Report.